

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

PRINCIPLES OF CELLULAR & MOBILE COMMUNICATIONS

(Common to EEE, CSE, IT, AI&DS, CSE (AI), CSE(DS), CSE(AI&ML), CSE(IOT), CSE(CS), CS&D and AI&ML)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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|---|----|
| (a) Why are cellular mobile communication systems necessary? | 2M |
| (b) How mobile radio communication evolved over time? | 2M |
| (c) What is cell splitting? | 2M |
| (d) What is the general description of the problem in cellular communication? | 2M |
| (e) What are the different types of handoffs? | 2M |
| (f) What is a soft handoff? | 2M |
| (g) What is Frequency Division Multiple Access (FDMA)? | 2M |
| (h) What is Time Division Multiple Access (TDMA)? | 2M |
| (i) What is the introduction to the 5G system? | 2M |
| (j) How does GSM handle data and voice communication? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Compare analog and digital cellular systems with at least 10 points each. 10M
- OR**
- 3 What is the uniqueness of mobile radio environment? Explain why hexagonal shaped cells are used for mobile communication. 10M
- 4 (a) For a seven cell reuse pattern find the reuse factor if the minimum distance between centers of co-channel-cells is 18 km. Radius of cell is 3 km and the distance between adjacent cells in the seven cell pattern is 6 km. 5M
- (b) What is cell-splitting? Explain its types in detail. 5M
- OR**
- 5 What are the different types of interference for a cellular system in detail? Also Explain the types of non-co-channel interferences in cellular system. 10M
- 6 (a) What are the various methods of delaying handoff? Explain briefly. 5M
- (b) What is meant by dropped call? Explain the factors that influenced dropped call rate. 5M
- OR**
- 7 Explain MAHO and soft handoff techniques. 10M
- 8 (a) Explain the architecture of TDMA. 5M
- (b) Explain in detail about multiple access schemes. 5M
- OR**
- 9 What is Space Division Multiple Access (SDMA), and how does it utilize spatial resources to manage multiple users in wireless communication? 10M
- 10 Compare the key characteristics and technologies used in 2G, 3G, and 4G systems. 10M
- OR**
- 11 Draw the GSM architecture and discuss various interfaces used in GSM. 10M
